Volume 2 is the one students underuse. Part 2 is a code cookbook organised by task — when you cannot remember how to build a ColumnTransformer or sweep a threshold, it is one lookup rather than a hunt through the chapters.

The twelve labs

Lab	Topic	Read first
0	**Preparation — do before Week 1**	—
1	Rows, joins, and what is missing	Ch 1–2
2	Summaries by hand, then in code	Ch 3
3	Clean the registry export	Ch 4
4	Pipelines and leakage	Ch 5
5	Exploratory analysis that survives scrutiny	Ch 6
6	Chart critique and redesign	Ch 7
7	Probability and simulation	Ch 8
8	Intervals, resampling, and power	Ch 9
9	The regression workflow	Ch 10
10	Classification, end to end	Ch 11–12
11	Clustering with honest uncertainty	Ch 13
12	Time series, text, and a dashboard	Ch 14–15

The setup cell

Put this at the top of every lab notebook. It never changes.

```
from google.colab import drive
drive.mount('/content/drive')

DATA = "/content/drive/MyDrive/fds/data/"    # change if your folder differs

import sys
```

```python
import numpy as np, pandas as pd, matplotlib.pyplot as plt
import sklearn, scipy

print("Python", sys.version.split()[0], "| pandas", pd.__version__,
    "| sklearn", sklearn.__version__)

students   = pd.read_csv(DATA + "students.csv")
enrolments = pd.read_csv(DATA + "enrolments.csv")
activity   = pd.read_csv(DATA + "activity.csv")
outcomes   = pd.read_csv(DATA + "outcomes.csv")

print(students.shape, enrolments.shape, activity.shape, outcomes.shape)
```

{=openxml}
<w:p><w:r><w:br w:type="page"/></w:r></w:p>

Lab 0 — Preparation

Do this before Week 1. About 20 minutes, on your own. Nothing here is data science — it is getting the tools working so that Lab 1 is not spent on setup.

If anything below fails, bring it to the first lab and someone will fix it in five minutes. Do not skip Lab 1 because your setup broke.

Step 1 — A Google account

Go to **https://colab.research.google.com** and sign in. Use your university account if you have one; a personal Gmail works.

Click **New notebook**. In the first cell type print("hello") and press **Shift + Enter**. If you see hello, Colab works.

Step 2 — Put the data in your Drive

Download the seven CSV files from the course page. Then in Google Drive:

1. Create a folder called fds
2. Inside it, create a folder called data
3. Upload all seven CSVs into fds/data/

Your path should end up as MyDrive/fds/data/students.csv. If you use a different path, you will need to change one line in every lab, so it is worth matching exactly.

Step 3 — Run the setup cell

In a new Colab notebook, paste this and run it:

```
from google.colab import drive
drive.mount('/content/drive')

DATA = "/content/drive/MyDrive/fds/data/"

import sys
import numpy as np, pandas as pd, matplotlib.pyplot as plt
import sklearn, scipy

print("Python", sys.version.split()[0], "| pandas", pd.__version__,
      "| sklearn", sklearn.__version__)

students   = pd.read_csv(DATA + "students.csv")
enrolments = pd.read_csv(DATA + "enrolments.csv")
activity   = pd.read_csv(DATA + "activity.csv")
outcomes   = pd.read_csv(DATA + "outcomes.csv")

print(students.shape, enrolments.shape, activity.shape, outcomes.shape)
```

A permission window will appear asking to connect Drive. Accept it. This is the one step that catches people — the window sometimes opens behind your browser.

You are ready when you see this:

```
(1200, 7) (4665, 5) (16165, 6) (1200, 7)
```

Step 4 — Learn the two menu items you will use every week

Runtime → Restart and run all. Cells run in the order *you* ran them, not top to bottom. This hides bugs that only appear when someone else opens your notebook. Run this before every submission.

File → Save a copy in Drive. Colab wipes its temporary storage after about 90 minutes of inactivity. Your Drive copy is safe; anything else is gone, with no recovery.

Step 5 — Read this before Lab 1

Read Chapter 1 of Volume 1. It is short. You do not need to understand all of it — you need to arrive knowing what a *unit of observation* is, because Lab 1 is about nothing else.

Troubleshooting

Problem	Fix
Drive mount does nothing	The permission window is behind your

Problem	Fix
	browser. Look for it.
FileNotFoundError	Your Drive path differs. Check MyDrive/fds/data/ exactly.
Shapes do not match	You uploaded the wrong files, or not all seven.
ModuleNotFoundError	Rare in Colab. Try !pip install packagename.

Lab 1 — Rows, joins, and what is missing

Reading: Chapters 1 and 2 · **Goal:** know what one row means before computing anything

0:00–0:10 · Setup

Run the setup cell from Lab 0. Confirm the four shapes.

If your setup did not work, sit next to someone whose did and get it fixed now. Do not spend the lab on this.

0:10–0:30 · Task 1 — What is a row?

For each of the four tables, write in a **Markdown cell**: “One row of this table is one _____.”

Then prove it:

```
for name, df in [("students", students), ("enrolments", enrolments),
                 ("activity", activity), ("outcomes", outcomes)]:
    print(f"{name:12s} {len(df):6d} rows, "
          f"{df.student_id.nunique():5d} distinct students")
```

Where a table has more rows than distinct students, name the extra dimension.

Then check whether every student appears the same number of times in activity:

```
print(activity.groupby("student_id").size().value_counts().sort_index().head())
```

They do **not**. Write one sentence proposing why. You do not have to be right — you have to notice.

0:30–0:55 · Task 2 — The count that is wrong

Compute the proportion of part-time students two ways:

```
merged = enrolments.merge(students[["student_id", "study_mode"]], on="student_id")
print("per registration:", (merged.study_mode == "Part-time").mean().round(3))
print("per student:     ", (students.study_mode == "Part-time").mean().round(3))
```

The answers differ, and one is wrong. In Markdown, say which and why, in two sentences.

This is the single most common error in applied data work. It produces a plausible number, raises no error, and survives every arithmetic check.

0:55–1:20 · Task 3 — Joins that break quietly

```
before = len(enrolments)
merged = enrolments.merge(students, on="student_id",
                          how="left", validate="many_to_one")
print(before, "→", len(merged))
```

validate="many_to_one" is the important part. Now break it on purpose:

```
broken = pd.concat([students, students.head(1)])  # one duplicate student
try:
   enrolments.merge(broken, on="student_id", how="left", validate="many_to_one")
except Exception as e:
   print("Caught:", e)
```

In Markdown: what would have happened without validate=? Use validate= on every merge from now on.

1:20–1:40 · Task 4 — Who is missing?

```
resp = outcomes.midterm_satisfaction.notna()
print(f"{resp.sum()} of {len(outcomes)} responded ({resp.mean():.1%})")
print(outcomes.groupby(resp).early_logins.mean().round(1))
```

Three-quarters responded, which sounds reassuring. Compare the two groups' engagement and write one sentence on what this does to any conclusion drawn from that survey.

1:40–1:50 · Checkpoint

Runtime → Restart and run all. Fix anything that breaks. Then **File → Save a copy in Drive**.

1:50–2:00 · Write-up

Write three questions this dataset could be asked: one **descriptive**, one **predictive**, and one it **cannot** answer — saying what evidence would be needed for the third.

Submit: your notebook.

> **Extension.** Compute mean logins per student two ways — activity.logins.mean(), and a per-student aggregate first. Predict the direction of the difference before running it, then explain the result using what one row means.